

Learning Seminar on Singularity Theory

Autumn 2023, AGMP group, Sheffield

1 Description

This learning seminar aims to be an introduction to some topics in singularity theory, in the case of (germ of) a holomorphic function $f : (\mathbb{C}^n, 0) \rightarrow (\mathbb{C}, 0)$ with an isolated critical point at origin. We will mostly focus on the study of Topology/Hodge Theory of families of varieties arising in this context - the family of nonsingular fibres of f or its miniversal deformation (usually called the Milnor Fibration). After a brief review on important notions from the Classification Theory of Singularities, we will discuss Picard Lefschetz Theory and Hodge Theory related to such a family. We will learn what are Vanishing Cycles, Monodromy Group, Gauss-Manin connection, Period map, Picard-Fuchs Equations, etc. As an application of these theories, we will learn one way to construct a Mixed Hodge Structure on the cohomology bundle.

We will also have two extra talks related to Singularity Theory and Mirror Symmetry. The first one would be about Saito Frobenius manifolds, which give certain flat structures on the miniversal deformation spaces. The second one would be about Homological Mirror Symmetry related to singularity theory, in which case vanishing cycles will take an important role in the A side category (Fukaya-Seidel Category).

For the main body of this seminar we will follow the guideline provided in Chapter I and Chapter II of [1], but not all topics in these two chapters will be introduced. A provisional plan with selected topics can be found in the next section.

Although [1] is a bit sketchy, in that book references are given for every important result. I also suggest the following books for references:

1. Supplement on Singularity Theory: [2] has more detailed explanations on most topics on [1, Chapter II]). Milnor has a book [6] about topology of Milnor fibres. See also [3] for a slightly different approach towards Singularity Theory.
2. Picard Lefschetz Theory: I found the last chapter of Liviu's book [7] quite friendly with enough details to learn how to prove the Picard-Lefschetz Formula.
3. Hodge Theory: Voisin's books [9] and [10] are classical.

2 Provisional Plan

As usual, each talk will be not more than **90 minutes**. Here is a provisional plan for the talks with a rough guide:

- (1) Equivalence between function germs and Versal Deformations

The first talk will be about an introduction to important notions from Classification Theory that will be used in our further discussions.

To Do: We select notions in [1, Chapter 1, §1] for the introduction. One possible way is, first explain what Equivalence and **Stable Equivalence** are, this allow us to compare functions with different number of variables. Then introduce the **Local Algebra** and explain what is the **Multiplicity** at a critical point. After that, we go to deformations of singularity. Introduce the notions of a **Versal Deformation** and **Infinitesimally Versal Deformation**. Then explain what is the **Level Bifurcation Set** (or Discriminant) of a singularity. Besides these notions, feel free to introduce other results in [1, Chapter 1, §1] you find interesting if time permits.

- (2) Picard Lefschetz Theory I

Picard Lefschetz Theory is the complex analogue of Morse theory. The difference is that when the function becomes holomorphic, the level manifolds close to a critical value will have the same topology. In Picard Lefschetz Theory, people study how a (co)homology class on a fiber changes as it follows the family of fibers of a whole circle around this critical value.

To Do: Following [1, Chapter2, 1.1-1.3] explain what is the **Homotopy Type of the Milnor Fibre** and how does **The Classical Monodromy Operator and Variation Operator** describe the behaviors of topology around a critical point. In the case of a Morse function, explain what is the **Vanishing Cycle** and how does **The Picard-Lefschetz Formula** describe the two operators we just mentioned. Use the function $f(z, w) = z^2 + w^2$ as an example as is explained on [1, Page 53].

(3) Picard Lefschetz Theory II

This is a continuation of the last talk. This time we will see how the discussions in Morse function case can be generalized to any isolated singularity. The idea is to replace f by an Morsification f_ϵ close to f , and then apply the results of Morse function.

To Do: Following [1, Chapter2, 1.4-1.6], first explain the local topology when we pass to a Morsefication. For a Morsefication, describe the **Vanishing Cycles**, explain how a distinguished system of paths determines a **Distinguished Basis** for homology of a generic fiber. Introduce the **Classical Monodromy Group** from the Morsefication point of view, and then introduce the **Intersection Matrix** and its relationship with the operators. Use A_3 singularity $f = z^4$ as an example, as is explained on [1, Page 60 and Page 63]. End up with discussions in [1, Chapter2, 1.10] about an alternative description of Monodromy Group that is independent of choice of Morsefication.

(4) Applied Hodge Theory I: Complex Monodromy and Period Maps

This talk and the next one will be about Hodge Theory in Singularity Theory. The Vanishing Cohomology Bundle - vector bundle over punctured disk/ complement of the Discriminant with fibers the DeRham cohomologies of Milnor fibers, will be our main character. The first talk will be about sections of the vanishing cohomology bundle.

To Do: Following [1, Chapter2, 3.1-3.6] introduce the **Vanishing Cohomology Bundle**, the **Gauss-Manin Connection** on it and how it defines the **Complex Monodromy Group**, then define the **Period Map** as a global section of the vanishing cohomology bundle, and in the context of singularity theory, introduce the **Residue Form** and **Geometric Sections**. Use a Morse singularity $f = z_1^2 + \dots + z_n^2$ to explain these notions. Explain results about the function Δ and how is it used to find a **Trivialization** of the Vanishing Cohomology Bundle.

(5) Applied Hodge Theory II: Differential Equations and Asymptotics of Integrals

This is the continuation of the last talk. The talk this time will be about differential equations from covariant derivative of the Gauss-Manin connection, the complex monodromy, asymptotics of the integrals and their relationship with classical monodromy operator.

To Do: Following [1, Chapter2, 3.7 and 3.8] recall what is the Complex Monodromy and explain how that is related to Gauss-Manin connection. Using the example on [1, Page 95] to explain how is the classical monodromy operator diagonalized. Introduce the **Picard-Fuchs Equation** coming from the Gauss-Manin connection and its property in the context of singularity theory. Explain the result on [1, Page 97] about relationships between asymptotic of integrals and classical monodromies.

(6) The Asymptotic Mixed Hodge Structure

On a compact Kähler manifold the Hodge Filtration defines a “pure” Hodge structure on its cohomology. For open varieties and singular varieties, such structure does not exist. Instead, the concept of a mixed Hodge structure is introduced. Using results from study of asymptotic of integrals, a mixed Hodge structure can be defined on the vanishing cohomology.

To Do: Following [1, Chapter2, 4.1-4.5] explain what is a **Pure Hodge Structure**, using the Hodge decomposition of a compact Kähler manifold as an example, and what is a **Mixed Hodge Structure**, using the complement of a codim 1 subvariety as an example. Introduce the Construction of the **Asymptotic Mixed Hodge Structure** on the cohomology bundle as described in the book.

(7) Simple Singularities

For simple singularities, our discussions in (2)-(5) admits a description in the corresponding root system R . The monodromy operator can be represented by a Coxeter transform in the Weyl group $W(R)$ and the period map extends to an isomorphism from the base of miniversal deformation to the orbit space of $W(R)$.

To Do: Explain the ADE classification of simple singularities ([1, Chapter1, 2.4]. Following [1, Chapter2, 5.1-5.6], introduce necessary concepts about root system and explain the first two theorems in 5.6. Use A_n singularities as an example.

(8) An introduction to Saito Frobenius Manifolds

The Saito Frobenius Manifolds give flat structures on the miniversal spaces of the isolated singularities. The tangent space of the deformation space is naturally equipped with a multiplication structure from the Local Algebra/Milnor Ring, and a symmetric bilinear form can be constructed from residues. When such a bilinear form is flat, (with a suitable Euler vector field) it gives a structure which is later called a Frobenius Manifold structure. A summary on this topic can be found in [5].

(9) Vanishing Cycles in Homological Mirror Symmetry

The Mirror Symmetry Conjecture predicts equivalence between B-Model theory of a variety X and A-Model theory of its mirror $\tilde{X}$. For a Fano variety X , the mirror will not just a variety- instead it will be a manifold with superpotential (a Landau-Ginzburg Model). At the level of categories, the Vanishing Cycles will take an important place on the A side of the Mirror family. This talk will be about Homological Mirror Symmetry related to Singularity Theory. Two possible topics under this subject are: HMS for Singularities (see [4]) and HMS for $\mathbb{C}P^2$ (see [8]). Feel free to choose other topics you like.

References

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